
What the Final Patch Hides: Event-Level Regression Measurement for Coding-Agent Harnesses

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Abstract

Coding agents are graded on the final patch alone. We instrumented an unmodified coding agent to replay the benchmark’s previously-passing tests after every edit, and found that transient regressions are common but almost never survive to the graded patch. Replacing our own harness’s model-written check with an execution-grounded one recovers the resolve that check was destroying (13 to 26 of 40; exploratory, and circular by construction). The instrument commits the working tree after every mutating tool call and replays the benchmark’s oracle at every commit inside its pinned container. We applied it to mini-swe-agent 2.4.6 unmodified, keeping its own loop, prompts, model layer, budgets, and submission protocol, and replacing only its environment object. On 40 SWE-bench Verified tasks it resolves 34 of 40 with `claude-sonnet-4-6` and 33 of 40 with `gpt-5.6-sol`, and every run ends with a changed tree. Eleven of the 80 broke previously-passing test functions, 135 counted per run and 100 distinct, one still failing when the official grader read the final tree. Its own testing does not close the gap. Over the window in which each test was broken, no test command ran the broken test in 6 of the 10 classifiable cases. In the other 4 one ran it and printed the failure by name. A method reading only the agent’s logs is structurally blind to most of them. The undercount holds in our harness under rollback (146 test functions broken across 8 of 47 runs, one still failing) and on SWE-bench Live. Two caveats we make measurable: rollback leaves clean trees partly because 17 of its 37 runs ship a tree they never changed, and a model-stack difference on Verified does not survive a change of scaffold. The instrument has its own blind spot. Nine of the ten trees that killed the suite recorded no event. We will release the instrument, its validation protocol, and the archived timelines. Appendix A catalogues 28 measurement defects met while building it, 23 of which produced plausible numbers rather than errors.

1 Introduction

Benchmarks for coding agents grade the final patch. SWE-bench [1] and its Verified subset [2] apply the submitted patch and report whether the target tests pass and the previously passing ones still pass, and work on agent-caused regressions counts what the submitted patch breaks [8]. That convention cannot see what happens while the agent works, because an agent that breaks a test at step three and repairs it at step five submits a clean patch. Two questions are left open: how often do agents break existing behaviour during a run, and how much survives?

The answer was smaller than we expected, and not in the place we expected. Agents break the oracle often and almost always repair it, so what the endpoint misses is transient and not shipped.

36 We measure the timeline instead of the endpoint. Every mutating tool call becomes a commit on a
37 git reference the agent cannot see, and the oracle is re-run at every commit inside the benchmark’s
38 pinned container. An agent operating system needs this observation primitive before it can build a
39 recovery primitive. We contribute:

- 40 1. **An instrument for event-level regression measurement**, applicable to an agent we did
41 not write. Every mutating tool call becomes a commit on a hidden git reference, and the
42 benchmark’s previously-passing oracle is replayed at every recorded state inside its pinned
43 container (Section 3).
- 44 2. **The measurement on an unmodified public scaffold**. mini-swe-agent 2.4.6 keeps its own
45 loop, prompts, model layer, budgets, and submission protocol. Only its environment object
46 is ours. It resolves 34 and 33 of 40 under two model families and breaks 135 oracle test
47 functions counted per run (100 distinct) across 11 of the 80 runs, one still failing at grade
48 time. In 6 of the 10 classifiable cases its own test commands never ran the test it broke
49 (Section 5.1).
- 50 3. **The same measurement across recovery policies**. The undercount holds in our harness
51 and on SWE-bench Live. But nearly half the rollback arm’s runs ship a tree they never
52 changed, so a clean final tree partly means no edits at all (Sections 5.2, 5.3, 5.5).
- 53 4. **Recovery policy depends on the monitor’s precision**. With a model-written check, rollback
54 buys 4 fewer contaminated trees for 11 fewer resolved instances (1 against 5; 13 against
55 24). Replacing that check with the repository’s own oracle gets both: 26 resolved, no
56 contaminated tree, and a lower cost per resolved instance than either. It is exploratory and
57 circular by construction (Section 5.4).
- 58 5. **A catalogue of 28 measurement defects** found while building the instrument (Appendix
59 A); 23 produced a plausible number rather than an error.

60 All measurements are exploratory, on development slices excluded from future confirmatory study.

61 2 Background and related work

62 **Benchmarks and their validity**. SWE-bench [1] grades a patch by tests that must go from failing
63 to passing and tests that must keep passing. Verified [2] is the human-screened subset, deprecated
64 as a capability metric [45]. Concerns about it have accumulated. They include contamination and
65 memorisation [6], plausible-but-wrong patches overstating resolve by 6.2 points [36], and insufficient
66 tests, which UTBoost [7] augmented to expose mislabelled patches affecting 24.4% of leaderboard
67 entries. Rolling [3, 5], private [4], and long-horizon [23] alternatives came in response. TDAD [8]
68 measured regressions in *submitted* patches on Verified (6.1% of previously-passing tests, 30.2% of
69 non-empty patches) and reduced them with an impact-analysis map. Process-level benchmarks score
70 intermediate states from logs [34, 35], and AgentLens [37] names regression cycles in 10.7% of
71 passing OpenHands runs, against our 11 of the 38 runs that edited substantially (29%). A log-derived
72 rate is bounded below by what the agent chose to run, and in 6 of our 10 classifiable cases it ran
73 nothing that would have executed the broken test. We execute the oracle at every mutating call
74 instead.

75 **Agent scaffolds and recovery**. SWE-agent [9], OpenHands [10], Agentless [11], and mini-swe-agent
76 [12] span the scaffold design space [13, 14]. Within-episode recovery appears as self-reflection [15,
77 16], progress-gated recovery [17], checkpoint repair [18], and aligned checkpoints [40]. Provenance-
78 based recovery is surveyed in [19]. Closest to us, Kim et al. [38] match logged edits against the
79 gold diff and recover agents that reach a gold-identical patch and then destroy it. Gao et al. [39]
80 bind verifier evidence to code states. Neither counts regressions or compares policies. Running
81 the repository’s tests during a run is deployed practice [42, 43]. Operating-system framings [20]
82 and the branch-context primitive [41] supply scheduling and fork, commit, and abort, but do not
83 observe the tree between actions. The regression problem is program repair’s plausible-versus-correct
84 patch problem [26, 28, 29], where regression tests are the main defence against overfitting [27].
85 Trajectory-level evaluation [21, 22] argues resolve rate alone is uninformative. Public leaderboard
86 trajectories record actions but not the tree at each step, which the instrument commits.

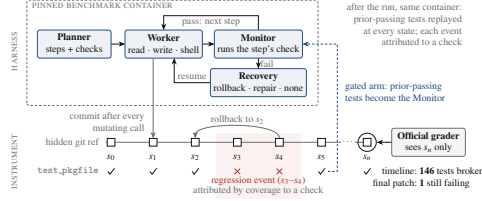


Figure 1: The harness and the instrument. Every mutating tool call commits the tree to a hidden git reference. After the run, the instance’s previously-passing tests are replayed at every state. The test row shows one regressing at s_3 – s_4 and repaired by a rollback to s_2 , which the official grader, reading only s_n , never sees. The counts are the GPT-5.6 rollback arm’s, over all 47 of its runs. In the gated arm (dashed), the timeline’s tests serve as the monitor.

3 Instrument

Figure 1 shows the harness and the instrument. **Observational timeline.** After every mutating tool call the working tree is committed to a git reference outside the agent’s view (a private index keeps its own `git status` unchanged). Rollbacks and the run’s end are observations too, and an archived run can be re-measured without re-running the agent. The agent’s tools and the harness’s checks execute inside the same pinned container as the replay, synchronised to the host tree the timeline records (Appendix A.2).

Exhaustive replay. For each observation the instance’s previously-passing (`PASS_TO_PASS`) tests run inside the pinned container against that observation’s tree. A regression event is a test that passes at one observation and fails at a later one. We replay every observation instead of bisecting, because a recovery policy makes verdicts non-monotone. Replay is scoped to the files holding those tests, which keeps it affordable on Verified (4 to 8 seconds per observation, about 26 CPU-minutes per sweep). Live’s larger oracles cost tens of minutes per instance. Infrastructure failures during replay, and tests already failing in the unmodified image, are recorded as missing observations, not failures.

Validation. Five checks run before any paid experiment (negative and positive controls, the latter including *recovered* regressions, a flake screen, an unknown-rate ceiling, a baseline liveness check), plus a golden check driving the gold patch through the agent’s real tool path and a re-scoring control that re-measures an archived timeline with no model calls. Re-scoring four arms’ bearing runs reproduced their episode counts and probe sets exactly. Appendix C gives the detail.

4 Experimental setup

Benchmark and slice. SWE-bench Verified is a measurement substrate here, not a leaderboard. Its pinned images and oracles make the measurement possible. A 40-instance development slice (16 from `django`), stratified and fixed before any measurement, was used throughout and is excluded from future confirmatory study. The GPT-5.6 stack also ran its first 10 as a calibration, and plain rollback ran twice. SWE-bench Live [3], built monthly from live repositories, is the second substrate, graded by its own harness whose rules we mirror. Its slice was fixed before any outcome (Appendix B) and its oracles are about twenty times larger (median 1,189 against 58 tests).

Harness. A planner decomposes the task into steps with verification commands, a worker executes each with three tools (`read`, `write`, `shell`), and a monitor runs the verification. On failure the recovery policy acts: **rollback** (reset to the last verified checkpoint and retry), **repair-in-place** (retry from the failed tree), or **no recovery** (the `FAIL` is final and the run ends). Runs have a \$4 work-cost cap. One instance under one policy is a cell. Each (stack, policy, substrate) triple is an arm.

Public scaffold. `mini-swe-agent 2.4.6` [12] runs unmodified (its own loop, prompts, model layer, 250-step and \$3 caps, and submission protocol) with only its environment object replaced, so every bash command executes in the benchmark’s pinned container and the tree is committed to the hidden reference after each. The seam is inert but not invisible. The agent’s `git diff` and `git status` show its own uncommitted edits and none of the instrument’s commits, but the synthetic baseline carries a marker name that 54 of the 80 runs saw, and we cannot settle here whether that changed behaviour. Six departures from its leaderboard configuration, a grid census putting its sampling inside

our arms’ range (a median of 3 and 2 tree-changing observations per run against our 2 to 5, at 9 to 15 bash commands each), and the rest are in Appendix C. We claim no equivalence to its published resolve rate, and because its budget differs in kind from our cap, no comparison between harnesses is tested.

Models. Two frontier stacks under identical harness, policy, instances and caps: `claude-opus-4-7/claude-sonnet-4-6` and `gpt-5.6-sol/gpt-5.6-terra` (planner/worker).

Outcomes. Resolve is the official grader’s verdict on the final patch. Events are reported at three levels because a few runs produce most: incidents (observations at which at least one oracle test broke, the co-primary endpoint), declared events (one per test function and onset, parametrised variants collapsed), and bearing runs (runs with at least one event). Incident *exposure* is a run’s incident count. An event and a final-state failure are different objects, so the undercount is in matched units (test functions broken during a run against those still failing at the end). Contamination counts oracle tests failing in the graded patch, net of baseline-dead. A tree that kills the suite is graded as failing every test but is a missing observation for event counting, so a contaminated cell need not be a bearing run.

Pre-declaration. The event unit, the proceed criteria, and the recovery-policy contrast were declared before the relevant data existed, and the analysis was committed before either comparison arm finished. One amendment preceded unblinding and one grading correction followed it (Section 5.3, Appendix B).

5 Results

5.1 An unmodified public scaffold breaks tests and puts them back

mini-swe-agent 2.4.6 ran unmodified on the same 40 instances. With `claude-sonnet-4-6`, 34 of 40 resolved (35 rot-aware) over 2,101 bash commands for \$31.38. Four runs hit its own \$3 cap. Three of those still grade resolved, because resolve reads the final tree, not the patch the scaffold never submitted. With `gpt-5.6-sol`, 33 of 40 resolved (34) over 937 commands for \$12.13. None of the 80 ended byte-identical to its start.

The timeline recorded 56 events in 5 of the 40 Claude runs and 79 in 6 of the 40 GPT-5.6 runs, on 8 instances, 3 bearing under both. Those 11 runs broke 135 oracle test functions counted per run (100 distinct, because the stacks overlap on the three shared instances), and when the grader read the final tree exactly one was still failing, `test_pkgfile` in `pytest-6197` (Claude 0 of 56, GPT-5.6 1 of 79). The scaffold has no monitor or rollback, so every closure is an ordinary forward edit. The eleven stayed open for 1 to 22 bash commands (median 4), and three runs supply 81% of the 135. Against exposure rather than attempts the rate is higher. No run making two or fewer tree-changing observations was bearing, and 11 of the 38 that made more broke something (29%). Much of what the endpoint misses is a transient inside one logical edit, but transients are not free: 14% of what these runs cost (\$1.29 of \$9.52) was spent while a regression was open.

One run in full. On `django-12965` the GPT-5.6 scaffold read `SQLDeleteCompiler` for twelve commands, then at command 13 changed `single_alias` to count active aliases with `<= 1`. Its own reproduction script crashed, and at command 15 `runtests.py delete` printed `ERROR: test_fast_delete_large_batch`. Command 16 replaced the edit, command 18 ran the suite clean, and the run graded resolved at 52 of 52. The timeline holds all of this. The graded patch holds none of it.

Would its own testing have caught it? For each breaking edit we read the trajectory over the window the test was broken, from the breaking command to just before the repair, asking whether a test runner ran, whether any run would have executed *this* test, and whether its output showed the failure. Coverage is three-valued, with *unknown* for runs that stopped early or were truncated (Appendix C). In 6 of the 10 classifiable cases the agent’s own test commands never ran the broken test. In the other 4 they ran it and printed the failure by name, and it was repaired anyway. The eleventh is unknown, a `pytest -x` run that stopped earlier. Self-testing is blind to the first group and ignored on the second. The endpoint sees neither. With eleven cases this is a case series, not a rate.

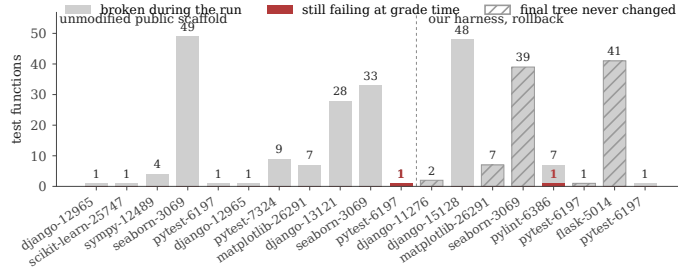


Figure 2: Every run that broke an oracle test, in matched units. Test functions broken on the timeline, dark where still failing at grade time. Left, the unmodified public scaffold under both models. Right, our rollback arm, hatched where the final tree is byte-identical to its start.

5.2 The same undercount in our harness

Under rollback the Claude stack resolved 13 of 40 (32.5%, exact 95% CI 18.6% to 49.1%) for \$34.24 and the GPT-5.6 stack 13 of 40 for \$8.14. A circuit breaker stopped three cells and a synchronisation defect lost two, so 37 were attempted and 35 graded (13 of 35, 37.1%). Rates are out of 40 unless stated, and one instance is unresolvable under the current parser (Appendix A.3), putting the ceiling at 39 for every arm, the scaffold included. The median rollback run makes 4 mutating tool calls under GPT-5.6 and 3 under Claude.

Eight of the 47 GPT-5.6 rollback runs (the sweep and a 10-instance calibration) broke a previously-passing test, 146 test functions counting each run separately (145 distinct). One of those, `test_can_read_toml_env_variable` in `pylint-6386`, was still failing at grade time. Three storms supply 88% of the 184 declared events over all 47 runs. Table 1’s 140 is the sweep alone, and declared events are exposure only. Figure 2 shows every bearing run on both sides.

A clean tree does not always represent work. Excluding the hidden reference, 17 of the 37 rollback runs end byte-identical to their start, as do 19 of the Claude stack’s 40, 4 under the gate, and 0 under repair-in-place. Four of the six bearing sweep runs shipped nothing. Over the two that delivered a patch, 1 of 55 broken test functions is still failing, so the claim rests on the public-scaffold arm, where no run ends unchanged.

Event frequency also differed by model stack, but does not generalise. Under the same harness and policy the Claude stack produced no events in 40 runs against the GPT-5.6 stack’s 140 in 6 of 37 (one-sided Fisher $p = 0.010$; reassign three bearing runs and $p = 0.11$), yet the same `claude-sonnet-4-6` worker under unmodified mini-swe-agent is bearing on 5 of 40 with 56 events, and the difference is absent on Live. Model and adapter are confounded, and controls are in Appendix C.

A break that stops the suite from being collected opens no timeline event, because a test the parser never mentions counts as a missing observation, not a failure, while official grading scores such a patch as failing everything. Nine of the ten suite-killed trees across all arms recorded no event, so the two instruments miss different things.

5.3 Recovery policy: what the clean final tree costs

On the primary endpoint, contamination paired by instance, no recovery was worse than rollback on 9 instances and better on 1 (exact sign test $p = 0.021$). Repair-in-place was worse on 5, better on 1 ($p = 0.22$). Incident exposure, the co-primary, did not differ ($p = 0.45$ and 1.0). The arms cannot show the policy caused the difference. No recovery ends at the monitor’s first FAIL, so its regressions arrive on the last tree-changing observation with nothing after them, and its 19-of-19 persistence is the stopping rule as much as the policy.

Rollback pays for its clean tree in delivered work, resolving 13 of 40 against 24 and 22, losing nine discordant pairs to repair-in-place and winning one (McNemar $p = 0.021$). The loss is the monitor’s. Of the 18 failed rollback runs, 15 failed at the first step, and 9 of those instances were resolved by an arm that kept work the check rejected and the grader accepted, and 17 of 37 runs ended unchanged.

Table 1: Every arm on both substrates. **empty**: runs ending byte-identical to their start. **events**: declared (test function, onset) pairs, exposure only, never a denominator. **broken** $\rightarrow$ **left**: over the bearing runs, oracle tests broken during the run against those still failing at grade time. **inc.**: incidents, the pre-declared co-primary. **contam.**: graded cells failing an oracle test, net of baseline-dead. The Claude rollback arm’s 1 is baseline rot, not agent damage (Appendix C). The public-scaffold rows ran under mini-swe-agent’s own budget, not our cap, so no comparison between harnesses is tested. Definitions, incidents, per-row composition and provenance: Appendix C.

substrate	arm	resolved	empty	events	inc.	bearing	broken $\rightarrow$ left	contam.	spend
Verified	mini-swe-agent, Claude	34	0/40	56	5	5/40	56 $\rightarrow$ 0	0	\$31.38
Verified	mini-swe-agent, GPT-5.6	33	0/40	79	6	6/40	79 $\rightarrow$ 1	1	\$12.13
Verified	GPT-5.6 rollback	13	17/37	140	11	6/37	104 $\rightarrow$ 1	1	\$8.14
Verified	Claude rollback	13	19/40	0	0	0/40	—	1	\$34.24
Verified	GPT-5.6 gated	26	4/40	70	7	5/40	62 $\rightarrow$ 0	0	\$10.51
Verified	GPT-5.6 split-gated	28	3/40	14	4	3/40	14 $\rightarrow$ 0	0	\$9.40
Verified	GPT-5.6 repair-in-place	24	0/40	84	7	7/40	83 $\rightarrow$ 4	5	\$14.73
Verified	GPT-5.6 no recovery	22	1/40	19	3	3/40	19 $\rightarrow$ 19	9	\$4.40
Live	GPT-5.6 rollback	0 (2)	25/39	156	9	6/39	147 $\rightarrow$ 0	0	\$10.29
Live	Claude rollback	2 (3)	24/39	49	6	3/39	48 $\rightarrow$ 0	1	\$49.80
Live	GPT-5.6 gated	1 (4)	9/39	280	6	3/39	270 $\rightarrow$ 0	2	\$11.49
Live	GPT-5.6 no recovery	4 (6)	3/39	30	4	4/39	30 $\rightarrow$ 30	4	\$8.95

212 **Disclosure.** The first unblinding gave $p = 0.375$, because seven cells whose final trees killed the suite
213 had been dropped as ungradable. Correcting the grader and re-grading those archived trees gives the
214 0.021 above, a rule change made after the data were seen (Appendix B).

215 5.4 Replacing the check removes the tradeoff

216 A fourth arm, added after unblinding (exploratory), replaces the planner-written check with the
217 repository’s own oracle, run after every attempt. A step is rejected only if a test that passed at the
218 start now fails. It resolved **26 of 40 (65.0%)**, above rollback’s 13 and the ungated arms’ 24 and 22,
219 below the 34 and 33 mini-swe-agent reaches under a budget differing in kind (Table 1), with **no**
220 **contaminated final tree** and 4 empty patches against rollback’s 17. The guarantee is cheap. The gate
221 runs at least 4 times in the median run at 4 to 8 seconds each, so 16 to 32 seconds of wall-clock and
222 \$2.37 more model spend than rollback, and per resolved instance it costs \$0.40 against rollback’s
223 \$0.63 and repair-in-place’s \$0.61. It beat rollback on 35 shared instances, winning 10 discordant
224 pairs and losing 1 ($p = 0.012$), and repair-in-place 26 to 24 with 0 contaminated trees against that
225 arm’s 5. The gate gives a guarantee, not a rate, and it is circular. Its tests are the benchmark’s own
226 oracle (median ratio 1.00), so a clean tree follows whenever the gate saw the final tree and the grader
227 reads what the gate read. A fifth arm weakens that, reading a deterministic half of the ids (2,278) and
228 never the other half (2,585), on which the grader also rules [44]. The split is over ids, not files, so a
229 break reaching a held-out id usually reaches a watched one, and the held-out half bounds the leak
230 instead of sampling it. A file-level or non-oracle gate would settle it. It resolved **28 of 40 (70.0%)**,
231 indistinguishable from the full gate ($p = 0.73$) and better than rollback (13 to 2, $p = 0.007$), with no
232 held-out failure apart from the parser-capped instance.

233 5.5 Replication on SWE-bench Live

234 Live keeps the undercount and loses the resolve rates, at most 6 of 40 even rot-aware, so its recovery
235 contrast is uninformative, and 25 of 39 rollback runs shipped an unchanged tree. Rollback broke 147
236 test functions across 6 of 39 runs, none still failing, against Verified’s 104 across 6. Live’s oracles are
237 twenty times larger, so the two counts are not comparable. No recovery left all 30 of its own ($p =$
238 0.125). The stack difference did not replicate (3 of 39 against 6, $p = 0.24$).

239 6 Discussion

240 **Why timeline regressions matter.** A repaired regression still costs money. Work done while one was
241 open took 14% of the scaffold’s bearing-run cost (\$1.29 of \$9.52) and 27% of our harness’s (\$0.63
242 of \$2.33), both small-base estimates. An endpoint-only benchmark also cannot tell a run that broke
243 nothing from one that broke 49 tests and repaired them. The events are the step-level ground truth
244 patch verifiers [24] and process reward models [25] need and instance-scale training sets [32, 33] do
245 not carry, and they tell a recovery primitive [41] when to abort. Two rules would have prevented most

of Appendix A’s defects: record infrastructure failures as missing observations, and require positive evidence that the measurement path is live. **Limitations.** Every rate estimates a 40-instance slice with one sweep per arm, and event counts bound the truth from both sides (Appendix C).

References

- [1] C. Jimenez et al. SWE-bench: Can language models resolve real-world GitHub issues? ICLR 2024. arXiv:2310.06770.
- [2] OpenAI. Introducing SWE-bench Verified. OpenAI blog, August 2024. openai.com/index/introducing-swe-bench-verified.
- [3] L. Zhang et al. SWE-bench Goes Live! NeurIPS 2025 Datasets and Benchmarks. arXiv:2505.23419.
- [4] X. Deng et al. SWE-Bench Pro: Can AI agents solve long-horizon software engineering tasks? arXiv:2509.16941, 2025.
- [5] I. Badertdinov et al. SWE-rebench: An automated pipeline for task collection and decontaminated evaluation of software engineering agents. NeurIPS 2025. arXiv:2505.20411.
- [6] S. Liang et al. The SWE-Bench Illusion: When state-of-the-art LLMs remember instead of reason. arXiv:2506.12286, 2025.
- [7] B. Yu et al. UTBoost: Rigorous evaluation of coding agents on SWE-Bench. ACL 2025. arXiv:2506.09289.
- [8] P. Alonso, S. Yovine, and V. A. Braberman. TDAD: Test-driven agentic development: Reducing code regressions in AI coding agents via graph-based impact analysis. arXiv:2603.17973, 2026.
- [9] J. Yang et al. SWE-agent: Agent-computer interfaces enable automated software engineering. NeurIPS 2024. arXiv:2405.15793.
- [10] X. Wang et al. OpenHands: An open platform for AI software developers as generalist agents. ICLR 2025. arXiv:2407.16741.
- [11] C. S. Xia et al. Agentless: Demystifying LLM-based software engineering agents. FSE 2025. arXiv:2407.01489.
- [12] K. Lieret and C. E. Jimenez. mini-swe-agent. Software, 2025. github.com/SWE-agent/mini-swe-agent.
- [13] X. Wang et al. Executable code actions elicit better LLM agents. ICML 2024. arXiv:2402.01030.
- [14] S. Yao et al. ReAct: Synergizing reasoning and acting in language models. ICLR 2023. arXiv:2210.03629.
- [15] N. Shinn et al. Reflexion: Language agents with verbal reinforcement learning. NeurIPS 2023. arXiv:2303.11366.
- [16] A. Madaan et al. Self-Refine: Iterative refinement with self-feedback. NeurIPS 2023. arXiv:2303.17651.
- [17] A. Kadu and A. Krishnan. ReflexGrad: Within-episode failure recovery in LLM agents via progress-gated dual-process routing. arXiv:2511.14584, 2025.
- [18] P. Mazaheri. REPOT: Recoverable program-of-thought via checkpoint repair. arXiv:2605.30052, 2026.
- [19] Y. Wang et al. From agent traces to trust: A survey of evidence tracing and execution provenance in LLM agents. arXiv:2606.04990, 2026.
- [20] K. Mei et al. AIOS: LLM agent operating system. COLM 2025. arXiv:2403.16971.

288 [21] S. Kapoor et al. Holistic Agent Leaderboard: The missing infrastructure for AI agent evaluation.
289 arXiv:2510.11977, 2025.

290 [22] R. Shu et al. What resolve rate hides: Trajectory structure diagnostics for coding agents.
291 arXiv:2607.06184, 2026.

292 [23] T. Le et al. SWE-EVO: Benchmarking coding agents in long-horizon software evolution
293 scenarios. arXiv:2512.18470, 2025.

294 [24] M. Raghavendra et al. Agentic rubrics as contextual verifiers for SWE agents. ACL 2026.
295 arXiv:2601.04171.

296 [25] M. L. Dihan and M. A. R. Khan. SWE-Shepherd: Advancing PRMs for reinforcing code agents.
297 arXiv:2604.10493, 2026.

298 [26] Z. Qi, F. Long, S. Achour, and M. Rinard. An analysis of patch plausibility and correctness for
299 generate-and-validate patch generation systems. ISSTA 2015. doi:10.1145/2771783.2771791.

300 [27] Y. Lou et al. When automated program repair meets regression testing: An extensive study on
301 two million patches. ACM TOSEM 33(7), 2024. arXiv:2105.07311.

302 [28] Z. Fei et al. Patch correctness assessment: A survey. ACM TOSEM 34(2), 2025.
303 doi:10.1145/3702972.

304 [29] H. Ye, M. Martinez, and M. Monperrus. Automated patch assessment for program repair at
305 scale. Empirical Software Engineering 26(2), 2021. doi:10.1007/s10664-020-09920-w.

306 [30] SWE-bench issue #601: Test result hijacking via stdout forging in evaluation harness.
307 github.com/SWE-bench/SWE-bench/issues/601, June 2026.

308 [31] OpenHands issues #4235 (October 2024) and #7044 (March 2025): Docker and environment
309 errors in SWE-bench instance evaluation. github.com/OpenHands/OpenHands.

310 [32] J. Yang et al. SWE-smith: Scaling data for software engineering agents. NeurIPS 2025 Datasets
311 and Benchmarks. arXiv:2504.21798.

312 [33] J. Pan et al. Training software engineering agents and verifiers with SWE-Gym. ICML 2025.
313 arXiv:2412.21139.

314 [34] M. B. Madiraju and M. S. P. Madiraju. RigorBench: Benchmarking engineering process
315 discipline in autonomous AI coding agents. arXiv:2606.22678, 2026.

316 [35] J. Chen et al. SWE-CI: Evaluating agent capabilities in maintaining codebases via continuous
317 integration. arXiv:2603.03823, 2026.

318 [36] Y. Wang, M. Pradel, and Z. Liu. Are "solved issues" in SWE-bench really solved correctly? An
319 empirical study. ICSE 2026. arXiv:2503.15223.

320 [37] P. Sahoo et al. AgentLens: Revealing the lucky pass problem in SWE-agent evaluation.
321 arXiv:2605.12925, 2026.

322 [38] M. Kim et al. Coherence collapse: Diagnosing why code agents fail after reaching the right
323 code. arXiv:2603.24631, 2026.

324 [39] X. Gao, J. Yang, and Q. Yang. Looping is not reliability: State-bound evidence and typed
325 revision contracts for agentic code repair. arXiv:2607.24604, 2026.

326 [40] Y. Zhuang et al. AgentRewind: Recoverable execution for long-horizon LLM agents.
327 arXiv:2608.14380, 2026.

328 [41] C. Wang and Y. Zheng. Fork, explore, commit: OS primitives for agentic exploration.
329 arXiv:2602.08199, 2026.

330 [42] Y. Chen, T. Ahmed, R. Jabbarvand, and M. Hirzel. Can old tests do new tricks for resolving
331 SWE issues? FSE 2026. arXiv:2510.18270.

- [43] P. Gao et al. Trae Agent: An LLM-based agent for software engineering with test-time scaling. arXiv:2507.23370, 2025.
- [44] T. Ahmed, J. Ganhotra, A. Shinnar, and M. Hirzel. Investigating test overfitting on SWE-bench. arXiv:2511.16858, 2025.
- [45] OpenAI. Why SWE-bench Verified no longer measures frontier coding capabilities. OpenAI blog, February 2026. openai.com/index/why-we-no-longer-evaluate-swe-bench-verified.

A The measurement defect catalogue

A.1 The catalogue by mechanism

Each row is a defect found while building the instrument. **S**: it produced a plausible number rather than an error. **G**: it was present while the test suite passed (A4 is the suite). **H**: it was findable only on a clean host. Class A rows depend on the machine the code runs on; class B rows render an infrastructure failure as a measurement; class C rows measure something other than the event as defined; class D rows consume a producer’s output without checking the producer.

#	Defect	Consequence	S	G	H
A1	Shadow commits inherited the machine’s git identity	timeline silently empty on any clean machine	✓	✓	✓
A2	Validation probe invoked bare <code>python</code>	every probe exit 127 on a clean host	✗	✓	✓
A3	Unpinned SDK resolved a different major version	two machines ran different code	✗	✓	✓
A4	Test fixtures verified with bare <code>pytest</code> from <code>PATH</code>	15 tests fail on a clean host as harness defects	✗	—	✓
A5	A benchmark scorer invoked bare <code>python</code>	score 0.0 from a missing interpreter	✓	✓	✓
A6	Agent executed on the host; measurement in the container	26 of 28 zero-step runs; see A.2	✓	✓	✓
B1	Output marker used shell :	a 13/13 run scored as 13 errors	✓	✓	
B2	Results on <code>stderr</code> , markers on <code>stdout</code>	one framework’s results outside the parsed slice	✓	✓	
B3	Probe diffed against a deleted commit	every graded test a hole	✓	✓	
B4	Interrupted mirror clone accepted with zero commits	later instances fail far from the cause	✓	✓	
B5	Container <code>workdir</code> unvalidated	every <code>exec</code> exit 127	✓	✓	
B6	Isolation removed loopback, not only the internet	23.5% of the oracle dead	✓	✓	
B7	Provider cache served removed containers	later cells replay against nothing and score clean	✓	✓	
B8	Tree reset reverted image build-time edits	one repository family’s oracle dead	✓	✓	
C1	Bisection assumed monotone verdicts	recovered regressions invisible	✓	✓	
C2	Declared event unit not implemented	rate off by up to $2.7\times$	✓	✓	
C3	Rollbacks produced no observation	recoveries invisible to the timeline	✓	✓	
C4	"Persists past the step boundary" undefined	one reading excludes the events of interest	✓	✓	
D1	Audit field never populated	0 tool errors reported, always	✓	✓	
D2	Malformed planner output crashed the run	33% of runs lost as task failures	✗	✓	
D3	Dependency install counted as agent work	attribution vacuous	✓	✓	
D4	Join took the first observation, not the last	failures pinned to the wrong tree	✓	✓	
D5	Executed config rebuilt from a name	manifest described a different run	✓	✓	
D6	Git handles never released	sweeps die after ~400 cells	✓	✓	
D7	Console re-walked the tree per run	presented as a dead server	✗	✓	
D8	Detection events lacked the ids attribution joined on	attributed detection always unknown	✓	✓	
D9	Absent coverage rendered as measured silence	unmeasured regressions reported as silent	✓	✓	
D10	Output-capped turns executed	half-written file; a 78-event storm	✓	✓	

Totals: 23 of 28 silent; 27 of 28 present under a passing suite; 6 of 28 findable only on a clean host. Ten further defects found after this census closed are described where they arose (Sections 5.2 and 5.3, Appendices A.3 and C); the rest will be released with the code.

A.2 The defect that invalidated a conclusion

After nineteen fixes, every validation check passed, the re-scoring control reproduced archived episode counts exactly, and three pilots (80 runs) measured an event rate far below the pre-declared proceed criterion. We drafted the conclusion the rule directed: the benchmark offered too little opportunity, so switch benchmarks. The conclusion was wrong. The harness ran the agent on the host

in a bare source checkout, while every probe and every gate ran inside the pinned container. On the host, `import matplotlib` in that checkout succeeds by importing the uncompiled source tree as a namespace package, and everything downstream fails in ways indistinguishable from an incompetent agent. Twenty-six of 28 zero-step runs traced to this. The checks had validated the measurement path and never the agent’s execution path. The fix was to route the agent’s tools and checks through the same container as the replay, and to add a per-cell environment parity check that runs before any model call.

361 A.3 Mechanisms in public infrastructure

362 Class A: OpenHands issues #4235 and #7044 [31], environment construction failures against SWE-
 363 bench images. Class B: UTBoost’s finding that the held-out oracle is insufficient at leaderboard scale
 364 [7], and, on SWE-bench Live, baseline tests that fail in the unmodified image because the calendar
 365 passed a deprecation date encoded in the instance. Class C: final-state regression measurement [8].
 366 Class D: the SWE-bench grader accepting forged test output on stdout [30], and, at the harness’s
 367 current revision, a parser that keys parametrised ids by their full text while the dataset stores them
 368 truncated at the first space (the earlier parser’s behaviour), so 16 previously-passing ids of one instance
 369 in our slice match no output and every submission to it grades as unresolved.

370 B Pre-declaration and analysis

371 The event unit is one (test function, onset observation) pair with parametrised variants collapsed. The
 372 gates for proceeding on a substrate were an event rate of at least 0.30 per run and a bearing fraction
 373 of at least 25%. Both stacks failed the conjunction (bearing 0% and 16.2%). We then re-declared the
 374 gates per (substrate, model stack) before the contrast was unblinded, so no arm here is confirmatory.
 375 The contrast’s primary endpoint is final-state contamination, paired by instance, exact two-sided
 376 sign test. The co-primary is incident exposure. The analysis script was committed before either
 377 comparison arm finished. The regression-gated arm was added after the contrast was unblinded and
 378 is exploratory. No multiplicity adjustment was planned or applied. Budget exhaustion is scored as
 379 failure. Instances whose baseline oracle records failures in the unmodified image are excluded from
 380 resolve-rate comparability claims, and their dead tests are typed as missing observations for event
 381 counting.

382 **Compute.** Every run, replay, and grade ran on one 12-vCPU, 22 GB x86 virtual machine running
 383 Docker, with no GPUs. The six harness arms on Verified in Table 1 cost \$81.42 in model spend and
 384 the two public-scaffold arms \$43.51, plus \$6.87 for the second rollback sweep. Calibration and pilots
 385 cost a comparable amount, and the Live sweeps report their own spend in Table 1.

386 **Slice construction on SWE-bench Live.** Fixed before any outcome: pytest-parsed instances with
 387 27 to 3,000 previously-passing tests, at most four per repository, earliest first, then the first 40 of 48
 388 candidates whose image passed a model-free environment check. No coverage maps are built for
 389 Live, so attribution there is unknown.

390 **Attribution (Verified only).** A detected regression is classified three ways: *attributed* if a failing
 391 harness check and the broken test both exercise a file the agent changed at that observation (per-
 392 instance coverage map at the base commit), *co-occurring* if some check failed while the regression
 393 was open without such a link, and *unknown* if the map cannot say. Of the calibration’s 57 raw
 394 episodes, 39 were attributed to a failing check, 16 co-occurred with one, and 2 were silent. Of the
 395 sweep’s 162, 48 had none. The public-scaffold arm has no monitor, so attribution is undefined and no
 396 silence statistic is reported.

397 **Grading correction between unblindings.** The first unblinding of the recovery-policy contrast gave
 398 $p = 0.375$ for the primary endpoint, because seven of the eight cells whose final trees made the suite
 399 uncollectable (five no-recovery, two repair-in-place) were dropped as ungradable. Official grading
 400 scores such a patch as failing every test, so the most contaminated states had been dropped from the
 401 endpoint they bore on most. The grader, not the replay, now distinguishes an environment failure
 402 from a patch that kills the suite, and we re-graded the seven archived trees without re-running any
 403 agent.

404 C The public-scaffold arm

405 **The seam.** mini-swe-agent’s loop touches the working tree in one place:
406 `DefaultAgent.execute_actions` calls `env.execute(action)`. We replace the environ-
407 ment object and nothing else, so the scaffold keeps its own loop, prompts, litellm model layer, step
408 and cost limits, and submission protocol. Our environment runs each bash command inside the
409 benchmark’s pinned container, pulls the resulting tree back to the host the timeline records, and
410 commits an observation when the tree changed. We measured two properties of the seam instead of
411 assuming them. First, the instrument does not advance the container’s git state, so the agent’s own
412 `git status`, `git log`, and `git diff` show its uncommitted work and none of the instrument’s
413 commits. The first paid pilot ran before this was true, and it spent eight of twenty commands
414 discovering that its edits had been committed underneath it. Second, the instrument is inert but
415 visible. The synthetic baseline commit it creates carries a fixed marker name, and 54 of the 80 runs
416 saw that name in some command’s output, 8 of them through `git status` (departure 2). We cannot
417 settle here whether seeing it changed behaviour. The 54 runs that saw it resolved 43 and were bearing
418 on 10, and the 26 that did not resolved 24 and were bearing on 1 (one-sided Fisher $p = 0.069$). A run
419 that edits more has both more to see and more to break, and that explains part of this, since no run
420 with two or fewer tree-changing observations was bearing at all under either condition. But within
421 the runs that edited more the association does not vanish (10 of 27 against 1 of 11, $p = 0.088$), so we
422 report it as an open question about the seam and not a settled confound. We also wrap the routed
423 command so that a command ending in a here-document terminator or a comment parses as the
424 scaffold’s own bash `-c` would parse it. Before that fix, 39 of 40 GPT-5.6 cells hit a shell syntax
425 error, because that model writes every script as a here-document.

426 **Departures from the scaffold’s leaderboard configuration.** (1) The measurement container has
427 no network, while its own deployment uses Docker’s default bridge. The unchanged prompt tells
428 the agent it may install tools, and here it cannot. (2) The container’s git history is a single synthetic
429 commit, because the image’s upstream history can contain the fix the task asks for. (3) Resolve is
430 graded on the final tree by the official grader, not on the scaffold’s submitted patch. (4) Our replay
431 and grading run outside its loop. (5) The two models are reached through the scaffold’s litellm layer,
432 GPT-5.6 through its `Responses-API` model class, which the `chat-completions` class cannot serve (it
433 returned three replies with no tool call and the scaffold exited on repeated format errors). (6) Prices
434 come from our verified table, because litellm’s carried a stale entry for one model. For these reasons
435 we claim no equivalence to its published resolve rate.

436 **Sampling grid.** Per run, median tree-changing observations and mutated files, with the run and final
437 observations excluded because neither is an opportunity to see a transition: public scaffold 3 and 3
438 (Claude), 2 and 2 (GPT-5.6); no recovery 2 and 2; repair-in-place 4 and 4; rollback 5 and 7; gate 4
439 and 4. Observations at which more than one file changed are 0 to 6% in every arm. The scaffold
440 issues 14.8 and 9.1 bash commands per observation against the harness’s one tool call per observation,
441 because most of its commands only read.

442 **The self-test classifier.** For each episode we take the window in which the test was broken (the
443 breaking command up to, but not including, the command that repaired it) and label three things over
444 the scaffold’s own trajectory. *Tested*: a test runner ran and its output proves it ran, so a runner that
445 died on an import does not count. *Covered*: one of those runs would have executed this test, decided
446 on the runner’s positional arguments matched at a label boundary (a run of two sibling modules does
447 not cover a test in a third), honouring `-k`, `-m`, `--deselect`, and explicit node ids. *Shown*: a covering
448 run’s output names this test on a line that also carries a failure marker. Coverage and display are
449 three-valued. A run that stopped at an earlier failure under `-x`, or whose output was truncated by a
450 pipe, is *unknown*, not negative, because calling it negative would assert that the agent ran the test and
451 was not told, from evidence that cannot support it. We audited the classifier’s labels by hand against
452 the raw trajectories of eight bearing cells from an earlier pilot of this arm, run before the two shell
453 defects below were fixed and so not among the runs reported here. The audit disagreed with an earlier
454 version of the classifier on all three totals, and the version reported here reproduces the audit exactly.

455 **The gate’s two contaminated Live trees.** One run hit the \$4 cap before the gate saw its final tree.
456 The other passed every check yet fails 13 grader tests, 7 in the file the hidden test patch rewrites.

457 **Uses.** The per-observation verdicts are the step-level ground truth that patch verifiers [24] and process
458 reward models [25] require. We release the archived timelines for that purpose.

459 **Further limitations.** Event counts are lower bounds in both directions: a break that stops the suite
460 from being collected opens no timeline event (Section 5.2), and no-recovery’s 19-of-19 persistence is
461 confounded with its stopping rule, since that arm ends at the monitor’s first FAIL. To separate policy
462 from truncation we would need a no-recovery arm that keeps executing after a failed check. The
463 public-scaffold arm is one scaffold at one version under two model families. Its budget differs in kind
464 from our arms’ (\$3 billed and 250 steps against a \$4 work cost), so the two harnesses are placed side
465 by side and never tested against each other. The self-test measurement rests on 11 breaking edits, one
466 of them coded unknown. Cross-stack comparisons confound model with adapter. The gated arms are
467 post hoc and select tests from benchmark metadata. Live resolve is near floor, and under rollback
468 most Live runs ship nothing, so its recovery contrast is uninformative on resolve.

469 **Bounding the model-stack difference (Section 5.2).** Exposure differs about twofold, not 140-fold.
470 In opportunity units the Claude arm makes 147 observations to the GPT arm’s 218, with per-run
471 medians of 3 observations and 3 mutated files against 5 and 7. The zero is measured, not a detection
472 failure. Re-scoring that stack’s earlier canary run reproduces its three events. A second plain-rollback
473 sweep reproduced the GPT side (13 of 36 graded resolved, 116 events, 6 of 37 bearing), so the
474 bearing fraction is stable even though per-instance event counts are not.

475 **The Claude rollback arm’s contaminated tree.** django-14238, a run whose tree never changed.
476 Grading its unmodified base tree three times returns 38 of 39 previously-passing tests each time. One
477 graded id never appears in the log, which upstream’s rule scores as a failure. It is baseline rot of the
478 kind Appendix A.3 describes, not a regression the timeline missed.

479 **The two readings.** The graded prediction excludes test files, as the official harness restores them
480 anyway, while the replay reads the tree as it stands, so an agent edit to a test file is visible to the
481 timeline and not to the grader.

482 **Reading Table 1.** *Baseline-dead* tests already fail in the unmodified image. *Rot-aware* resolve
483 counts a run resolved when every target test passes and no failure lies outside that set. On Live,
484 **resolved** is of 40 while **empty** and **bearing** are of the 39 attempted (torch-tune-1806 exceeds the
485 replay budget). **broken** → **left** omits any cell whose suite the grader could not collect, since such a
486 cell would otherwise read as perfect capture, so repair-in-place’s row covers six of its seven bearing
487 runs and 83 of its 84 events.

488 **Incidents.** The pre-declared co-primary, per arm, in the order of Table 1: 5, 6, 11, 0, 7, 4, 7, 3, 9, 6, 6,
489 4.

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Justification: No models are trained. Section 4 gives the slices and how they were fixed, the harness and its three recovery policies (the two regression-gated variants are specified in Section 5.4), the public scaffold and its unmodified budgets, the exact model strings, the per-run cost cap, the outcome definitions, and the pre-declared analysis.

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The Claude rollback arm’s resolve carries an exact (Clopper–Pearson) 95% interval; paired comparisons use exact sign tests and exact McNemar tests, and the cross-stack comparison a one-sided Fisher exact test. The tests reported and the absence of multiplicity adjustment are declared (Section 5.4, Appendix B). Plain rollback was swept twice on Verified (Section 4) but the replicate’s numbers are not reported here, and intervals treat instances as independent although 16 of the 40 Verified instances come from django.

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Justification: The contribution is an evaluation instrument and measurements of it, and the plausible direct impact is positive: agents whose regressions are visible break less existing behaviour, and the archived timelines give patch verifiers and process reward models step-level supervision they currently have to guess at. Two negative paths are worth naming. An instrument that commits the working tree after every action is agent telemetry: deployed outside a benchmark it would record whatever an agent touches, including proprietary or personal code, so the hidden reference is a surveillance surface as much as a measurement one. And a gate that rejects any step failing a prior-passing test is an enforcement mechanism, not only an observer: applied to real repositories it would suppress legitimate refactors that temporarily break tests, which is the behaviour a maintainer sometimes wants. Both argue for the gate being advisory and the timeline being local and short-lived by default.

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